PA

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DSA Prep Series

Placement Prep Edition

• PLACEMENT PREP SERIES

100 DSA Coding Questions

The most **frequently asked** Data Structures & Algorithms questions in campus placement and assessment rounds — **compiled from real candidate experiences, 2021 to 2026.**

- Arrays & Strings
- Trees & Graphs
- Dynamic Programming
- + 8 more topics

21-26

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DSA Prep Series

• STUDY ROADMAP

Your 10-Step DSA Roadmap

Follow this order before attempting the question bank on the next pages — it mirrors the sequence most assessment rounds build on.

01 Arrays & Strings
Traversal, two-pointer technique, sliding window basics

02 Searching & Sorting
Binary search, merge sort, quicksort fundamentals

03 Linked Lists
Reversal, cycle detection, merging two lists

04 Stacks & Queues
Monotonic stacks, queue simulation, expression evaluation

05 Hashing & Maps
Frequency counting, fast lookups, pair problems

06 Recursion & Backtracking
Base cases, state-space search, pruning

07 Trees & BSTs
Traversals, height, ancestors, balancing basics

08 Graphs
BFS / DFS, shortest paths, connected components

09 Dynamic Programming
Memoization, tabulation, classic DP patterns

10 Greedy & Bit Manipulation
Interval scheduling, XOR tricks, bit counting

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TOPIC 01 / 12

Arrays

Q01 Find the missing number in an array containing numbers from 1 to N.

Q02 Move all zeros in an array to the end without changing the order of other elements.

Q03 Find two numbers in an array that add up to a given target sum.

Q04 Find the maximum sum of a contiguous subarray (Kadane's Algorithm).

Q05 Rotate an array to the right by K positions.

Q06 Find the duplicate number in an array of N+1 integers.

Arrays — continued

Q07 Find the intersection of two sorted arrays.

Q08 Find the equilibrium index where the left sum equals the right sum.

Q09 Merge two sorted arrays into one sorted array in place.

Q10 Find the majority element that appears more than N/2 times.

Q11 Find the maximum product of a contiguous subarray.

Q12 Rearrange array elements alternately as positive and negative numbers.

TOPIC 02 / 12

Strings

Q13 Check whether a given string is a palindrome.

Q14 Check if two strings are anagrams of each other.

Q15 Find the first non-repeating character in a string.

Q16 Reverse the words in a sentence without reversing individual words.

Q17 Find the longest substring without repeating characters.

Strings — continued

Q18 Count the number of vowels and consonants in a string.

Q19 Implement basic string compression, e.g. aabccccccaaa becomes a2b1c5a3.

Q20 Check if a string can be rearranged into a palindrome.

Q21 Print all permutations of a given string.

Q22 Find the longest common prefix among an array of strings.

TOPIC 03 / 12

Linked List

Q23 Reverse a singly linked list.

Q24 Detect a cycle in a linked list.

Q25 Find the middle node of a linked list in a single pass.

Q26 Merge two sorted linked lists into one sorted list.

Linked List — continued

Q27 Remove the Nth node from the end of a linked list.

Q28 Detect and remove a loop in a linked list.

Q29 Check whether a linked list is a palindrome.

Q30 Find the intersection point of two linked lists.

TOPIC 04 / 12

Stacks & Queues

Q31 Check for balanced parentheses in an expression using a stack.

Q32 Implement a queue using two stacks.

Q33 Design a stack that returns the minimum element in O(1) time.

Q34 Evaluate a postfix expression using a stack.

Stacks & Queues — continued

Q35 Find the next greater element for every item in an array.

Q36 Implement a circular queue with a fixed size.

Q37 Sort a stack using only stack operations, without an extra array.

TOPIC 05 / 12

Trees

Q38 Find the height of a binary tree.

Q39 Perform a level order traversal (BFS) of a binary tree.

Q40 Check whether a binary tree is a valid binary search tree.

Q41 Find the lowest common ancestor of two nodes in a binary tree.

Q42 Convert a binary tree into its mirror image.

Trees — continued

Q43 Find the diameter of a binary tree.

Q44 Check if two binary trees are structurally identical.

Q45 Print the boundary nodes of a binary tree.

Q46 Find the Kth smallest element in a binary search tree.

Q47 Serialize and deserialize a binary tree.

TOPIC 06 / 12

Graphs

Q48 Implement a breadth-first search (BFS) traversal of a graph.

Q49 Implement a depth-first search (DFS) traversal of a graph.

Q50 Detect a cycle in a directed graph.

Q51 Find the shortest path between two nodes in an unweighted graph.

Q52 Count the number of connected components in a graph.

Graphs — continued

Q53 Check whether a graph is bipartite.

Q54 Find the number of islands in a 2D grid.

Q55 Implement topological sorting of a directed acyclic graph.

Q56 Check if a path exists between two nodes using Union-Find.

TOPIC 07 / 12

Recursion & Backtracking

Q57 Generate all subsets (the power set) of a given array.

Q58 Solve the N-Queens problem.

Q59 Generate all valid combinations of balanced parentheses.

Q60 Solve a Sudoku puzzle using backtracking.

Recursion & Backtracking — continued

Q61 Find all possible paths in a maze from source to destination.

Q62 Generate all permutations of a given array.

Q63 Solve the classic Rat in a Maze problem.

Q64 Compute the power of a number using fast recursive exponentiation.

TOPIC 08 / 12

Dynamic Programming

Q65 Solve the 0/1 Knapsack problem.

Q66 Find the longest common subsequence between two strings.

Q67 Find the minimum number of coins needed to make a given amount.

Q68 Solve the climbing stairs problem by counting distinct ways.

Q69 Find the longest increasing subsequence in an array.

Q70 Solve the edit distance problem between two strings.

Dynamic Programming — continued

Q71 Find the maximum sum of non-adjacent elements (House Robber).

Q72 Solve the matrix chain multiplication problem.

Q73 Count the number of unique paths in a grid from top-left to bottom-right.

Q74 Solve the subset sum problem.

Q75 Find the longest palindromic substring.

Q76 Solve the partition equal subset sum problem.

TOPIC 09 / 12

Searching & Sorting

Q77 Implement binary search on a sorted array.

Q78 Find the first and last occurrence of an element in a sorted array.

Q79 Implement merge sort and explain its time complexity.

Q80 Implement quicksort and explain the partitioning logic.

Searching & Sorting — continued

Q81 Search for an element in a rotated sorted array.

Q82 Find the Kth largest element in an unsorted array.

Q83 Sort an array of only 0s, 1s and 2s (Dutch National Flag problem).

Q84 Find a peak element in an array.

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TOPIC 10 / 12

Greedy

Q85 Solve the activity selection / job scheduling problem.

Q86 Find the minimum number of platforms needed at a railway station.

Q87 Solve the fractional knapsack problem.

Q88 Find the minimum coins needed using the greedy approach.

Q89 Solve the meeting rooms problem to find overlapping intervals.

Q90 Assign cookies to children to maximize the number satisfied.

TOPIC 11 / 12

Hashing & Maps

Q91 Find the first repeating element in an array.

Q92 Group anagrams together from a list of strings.

Q93 Find all pairs in an array with a given difference.

Q94 Check whether a subarray with sum zero exists.

Q95 Find the longest consecutive sequence in an unsorted array.

Q96 Count the distinct elements in every window of size K.

TOPIC 12 / 12

Bit Manipulation

Q97 Find the single number that doesn't repeat when every other number repeats twice.

Q98 Count the number of set bits in an integer.

Q99 Check whether a given number is a power of two.

Q100 Find the XOR of all numbers from 1 to N.

• YOU MADE IT

Now go practice.

Save this PDF, revisit one topic a day, and pair every question here with a whiteboard walkthrough — not just a mental read-through.

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